



Unleash your potential

Module Presenter's Manual

for

Developing Applications with Python

***Effective From: June 2024
Ver. 1.0***

Amendment Record

Version No.	Effective Date	Change	Replaced Pages
1.0	June 2024	New	-

Table of Contents

Sr. No.	Details	Page No.
1.	Introduction	1
2.	Information on Session Allocation	1
3.	Session Deliverables	2
4.	Week-wise Session Schedule	3
5.	Session Coverage	4
6.	Library References	11

1. Introduction

At the end of this course, students will be able to:

- Learn the concept of Python programming
- Working with Functions and Function Parameters in Python
- Understand the benefits of Iterators, Generators, and Decorators
- Learn how to use Serialization and Deserialization

2. Information on Session Allocation

Module	Online Hours
Developing Applications with Python	40

Throughout this Presenter's Manual, the module *Developing Applications with Python* will be referred to as **DAPYT**.

3. Session Deliverables

To aid the teaching process, following are the deliverables

(Faculties can access eBooks and other components in Onlinevarsity through Aptrack.)

Resources available on Onlinevarsity:

Icons	Feature - Description/Functionality
	Download Book - Student has the option to download the subject related e-book and read offline.
	Practice 4 Me - Student can test and evaluate their understanding of module related topics.
	Work Assignments - Student can solve scenario based lab assignments (Hands-on). The faculty will evaluate and give their feedbacks.
	References - Student can access additional subject related material for reading.
	Feedback - Student can provide feedback on the course material.
	Ask to Learn – Student can submit subject related technical queries. Queries submitted will be directed to the particular course coordinator/head.

4. Week-wise Session Schedule

- A Session is of 2 hours duration.

➤ Week-Wise Schedule

Week	Day 1	Day 2	Day 3	Day 4
1	Session 1 DAPYT – TL1	Session 2 DAPYT – TL2	Session 3 DAPYT – TL3	Session 4 DAPYT – TL4
2	Session 5 DAPYT – TL5	Session 6 DAPYT – TL6	Session 7 DAPYT – TL7	Session 8 DAPYT – TL8
3	Session 9 DAPYT – TL9	Session 10 DAPYT – TL10	Session 11 DAPYT – TL11	Session 12 DAPYT – TL12
4	Session 13 DAPYT – TL13	Session 14 DAPYT – TL14	Session 15 DAPYT – TL15	Session 16 DAPYT – TL16
5	Session 17 DAPYT – TL17	Session 18 DAPYT – TL18	Session 19 DAPYT – TL19	Session 20 DAPYT – TL20

DAPYT: Developing Applications with Python
TL: Online Session

5. Session Coverage

Session No.	Session Title	Session Details	Deliverables' Mapping
1	DAPYT – TL1	All the topics as listed below from Session 1 and Session 2 of <i>Learning Python By Example</i> book should be covered in this session. <u>Session 1 – Introduction to Python</u> ➤ Explain the concept of Python Programming ➤ Explain briefly the history of Python ➤ List the features, usage, and benefits of Python ➤ Outline the process to create a Python Program ➤ Explain about identifiers ➤ Identify and list the keywords in Python <u>Session 2 – Data Types and Operators in Python</u> ➤ Explain the data types in Python ➤ Distinguish between different numeric types ➤ Explain type conversion functions in Python ➤ Categorize different operators in Python ➤ Explain about identifiers ➤ Identify the keywords in Python	Learning Python By Example SG – Session 1 & 2 XP – Session 1 & 2 TG – Session 1 & 2
2	DAPYT – TL2	All the topics as listed below from Session 3 and Session 4 of <i>Learning Python By Example</i> book should be covered in this session. <u>Session 3 – Flow Control Statements in Python</u> ➤ Define the use of control flow and its importance ➤ Describe the <code>if</code>, <code>elif</code>, and <code>else</code> statements ➤ Identify the best practices to maintain code readability	Learning Python By Example SG – Session 3 & 4 XP – Session 3 & 4 TG – Session 3 & 4

Session No.	Session Title	Session Details	Deliverables' Mapping
		when using nested conditionals ➤ Explain the <code>while</code> and <code>for</code> loop and its usage for iterating over sequences ➤ Distinguish <code>break</code> and <code>continue</code> statements and their impact on the loop execution ➤ Identify process of exception handling <u>Session 4 – Functions and Function Parameters in Python</u> ➤ Define the basics of functions in Python ➤ Explain the types of arguments and concept of parameters ➤ Identify the <code>user-defined</code> functions in Python ➤ Explain <code>lambda</code> functions and <code>callables</code> in Python	
3	DAPYT – TL3	The Try It Yourself Questions from Session 1 to Session 4 of <i>Learning Python By Example</i> book should be covered in this session.	Learning Python By Example (Onlinevarsity) Session 1-4
4	DAPYT – TL4	All the topics as listed below from Session 5 of <i>Learning Python By Example</i> book should be covered in this session. <u>Session 5 – List and Tuples in Python</u> ➤ Outline the concept of List in Python ➤ Explain the creation and working of Lists in Python ➤ Describe the process of receiving input and adding elements ➤ Define the process of concatenating and slicing Tuples in Python	Learning Python By Example SG – Session 5 XP – Session 5 TG – Session 5

Session No.	Session Title	Session Details	Deliverables' Mapping
5	DAPYT – TL5	All the topics as listed below from Session 6 of <i>Learning Python By Example</i> book should be covered in this session. <u>Session 6 – Dictionaries and Sets in Python</u> ➤ Describe dictionaries and key-value pairs ➤ Explain working with dictionaries operations ➤ Distinguish between shallow and deep copies of dictionaries ➤ Define sets using the <code>set()</code> constructor or curly braces <code>{}</code> ➤ Explain about set operations and set methods	Learning Python By Example SG – Session 6 XP – Session 6 TG – Session 6
6	DAPYT – TL6	The Try It Yourself Questions from Session 5 and Session 6 of <i>Learning Python By Example</i> book should be covered in this session.	Learning Python By Example (Onlinevarsity) Session 5 & 6
7	DAPYT – TL7	All the topics as listed below from Session 7 and Session 8 of <i>Learning Python By Example</i> book should be covered in this session. <u>Session 7 – Iterators, Generators, and Decorators</u> ➤ Define an iterator and its purpose in Python programming ➤ Explain the benefits of iterators ➤ Explain the iterator protocol ➤ Describe the iterability of common Python data structures ➤ Explain the use of for loop to iterate over various types of iterables ➤ Distinguish between regular functions and generator functions ➤ Describe the memory-efficient iteration	Learning Python By Example SG – Session 7 XP – Session 7 TG – Session 7

Session No.	Session Title	Session Details	Deliverables' Mapping
		➤ Define a decorator and its roles ➤ Explain the applications of decorators ➤ Describe the creation of decorators using classes	
8	DAPYT – TL8	All the topics as listed below from Session 8 of <i>Learning Python By Example</i> book should be covered in this session. <u>Session 8 – Object Oriented Programming (OOPS)</u> ➤ Describe a class in Python ➤ Explain the use of objects and Constructor ➤ List the Methods, Getter, and Setter ➤ Explain the use of Static Field ➤ Define and describe Garbage Collection	Learning Python By Example SG – Session 8 XP – Session 8 TG – Session 8
9	DAPYT – TL9	All the topics as listed below from Session 9 of <i>Learning Python By Example</i> book should be covered in this session. <u>Session 9 – Advanced OOPS and Exceptional Handling</u> ➤ Explain the concept of Encapsulation ➤ Explain briefly about Inheritance ➤ Outline the concept of Polymorphism	Learning Python By Example SG – Session 9 XP – Session 9 TG – Session 9
10	DAPYT – TL10	All the topics as listed below from Session 9 of <i>Learning Python By Example</i> book should be covered in this session. <u>Session 9 – Advanced OOPS and Exceptional Handling</u> ➤ Describe about Abstraction in Python ➤ Identify Exception Handling ➤ Describe the concepts of Assertions and Logging in Python	Learning Python By Example SG – Session 9 XP – Session 9 TG – Session 9

Session No.	Session Title	Session Details	Deliverables' Mapping
11	DAPYT – TL11	The Try It Yourself Questions from Session 7 to Session 9 of <i>Learning Python By Example</i> book should be covered in this session.	Learning Python By Example (Onlinevarsity) Session 7-9
12	DAPYT – TL12	All the topics as listed below from Session 10 of <i>Learning Python By Example</i> book should be covered in this session. <u>Session 10 – Serialization and Deserialization</u> ➤ Explain the concept of serialization in Python ➤ Describe the purpose of deserialization in Python ➤ Distinguish between serialization and deserialization processes ➤ Identify scenarios where serialization and deserialization are used in Python ➤ Explain the importance of exceptions and error handling in serialization and deserialization	Learning Python By Example SG – Session 10 XP – Session 10 TG – Session 10
13	DAPYT – TL13	The Try It Yourself Questions from Session 10 of <i>Learning Python By Example</i> book should be covered in this session.	Learning Python By Example (Onlinevarsity) Session 10
14	DAPYT – TL14	All the topics as listed below from Session 11 of <i>Learning Python By Example</i> book should be covered in this session. <u>Session 11 – Files and Threads</u> ➤ Define File in Python programming ➤ Identify steps involved in reading and writing string in a File ➤ Explain the concept of pickle in Python	Learning Python By Example SG – Session 11 XP – Session 11 TG – Session 11

Session No.	Session Title	Session Details	Deliverables' Mapping
15	DAPYT – TL15	All the topics as listed below from Session 11 of <i>Learning Python By Example</i> book should be covered in this session. <u>Session 11 – Files and Threads</u> ➤ Describe creating a Thread using both a Class and a Function ➤ Explain the concept of Multithreading in Python ➤ Identify the utilization of Thread in communication	Learning Python By Example SG – Session 11 XP – Session 11 TG – Session 11
16	DAPYT – TL16	The Try It Yourself Questions from Session 11 of <i>Learning Python By Example</i> book should be covered in this session.	Learning Python By Example (Onlinevarsity) Session 11
17	DAPYT – TL17	All the topics as listed below from Session 12 of <i>Learning Python By Example</i> book should be covered in this session. <u>Session 12 – Network and Database Handling</u> ➤ Identify steps involved in accessing HTML, Images, and Server ➤ Explain how to install and setup MySQL Database	Learning Python By Example SG – Session 12 XP – Session 12 TG – Session 12
18	DAPYT – TL18	All the topics as listed below from Session 12 of <i>Learning Python By Example</i> book should be covered in this session. <u>Session 12 – Network and Database Handling</u> ➤ Explain how to create a Database in Python ➤ Define CREATE, READ, UPDATE and DELETE commands	Learning Python By Example SG – Session 12 XP – Session 12 TG – Session 12
19	DAPYT – TL19	All the topics as listed below from Session 12 of <i>Learning Python By</i>	Learning Python By Example

Session No.	Session Title	Session Details	Deliverables' Mapping
		<i>Example</i> book should be covered in this session. <u>Session 12 – Network and Database Handling</u> ➤ Describe how to download and use tools in Python	SG – Session 12 XP – Session 12 TG - Session 12
20	DAPYT – TL20	The Try It Yourself Questions from Session 12 of <i>Learning Python By Example</i> book should be covered in this session.	Learning Python By Example (Onlinevarsity) Session 12

6. Library References

- | |
|--|
| ➤ Python by Example: Learning to Program in 150 Challenges by Nichola Lacey |
| ➤ Python Programming 2E by Reema Thareja |
| ➤ Learning Python by Fabrizio Romano |

~~~ End of Document ~~~